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Parameter-efficient deep-learning-based model for segmentation with radiomic feature extraction

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ABSTRACT. **Purpose:** Magnetic resonance imaging (MRI), particularly dynamic contrast-enhanced MRI (DCE-MRI), plays a vital role in breast cancer assessment by highlighting tumor regions. Accurate 3D segmentation of tumors can significantly aid in diagnosis, disease monitoring, and treatment planning. Current state-of-the-art models such as nnU-Net are computationally expensive, with high parameter counts and memory requirements. In this work, we propose a parameter-efficient convolutional neural network-based architecture tailored for breast tumor segmentation in DCE-MRI.

Approach: The model integrates lightweight residual blocks into the SegResNet backbone and is trained on the first 3 DCE-MRI phases. We test the addition of the FRLoss from the HCMA-UNet model in the FRLoss ablation. Its encoder–decoder design also enables exploration of the treatment response prediction to neoadjuvant chemotherapy measured by the pathological complete response, using the output of the last encoder block. The last encoder block output is average-pooled and used as input to an XGBoost model with two estimators (max depth 5, learning rate 1).

Results: Evaluated on the public MAMA-MIA dataset, our proposed model achieves comparable performance to nnU-Net and SegResNet, with a 0.99% higher Dice score than nnU-Net, while reducing parameter count by 91.5%, FLOPs by 85.05%, and memory usage by 31.94% compared with nnU-Net. Therefore, the proposed model is significantly more efficient than nnU-Net and also offers superior accuracy to the Mamba-based baseline, even though the Mamba baseline remains computationally lighter demonstrated by its faster inference speed and lower giga floating point operations per second (38 versus 87). The XGBoost model for treatment response prediction does not demonstrate competitive performance with a balanced accuracy of 57.2% and receiver operating characteristic–area under curve of 0.542.

Conclusion: Our results demonstrate that parameter-efficient models can achieve competitive performance in DCE-MRI tumor segmentation.

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Keywords: dynamic contrast-enhanced imaging; 3D segmentation; parameter-efficient models; treatment response

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1 Introduction

Magnetic resonance imaging (MRI) is widely used in breast cancer assessment because of its high sensitivity as an imaging modality.[1] Dynamic contrast-enhanced MRI (DCE-MRI) uses contrast

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agents to enhance blood vessels and tissues, improving tumor localization.[1,2] Accurate tumor characterization (e.g., shape and volume) aids in tracking disease progression and treatment responses,[1,3] and a model that can provide gold standard segmentations can assist in the development of artificial intelligence (AI) models for diagnosis and prognosis.[1] Segmentations are often used to denote the region in the original MRI image to extract radiomic features.[4] These radiomic features can be used to predict treatment responses.[4] However, 3D medical segmentation is computationally expensive; state-of-the-art (SOTA) models such as nnU-Net[5] use over 30 M parameters, high FLOPs, and substantial memory. Segmentation of breast tumors is a hard problem due to the relatively small size of tumors compared with the total 3D volume. DCE-MRI helps emphasize tumor regions, suggesting that lightweight models could be effective for this task.

The proposed architecture contains lightweight residual blocks (ResBlocks) and uses the cross-entropy (CE)[6] and Dice[7] loss. We evaluated the proposed model on the public MAMA-MIA dataset[1] and achieved comparable performance to nnU-Net and SegResNet,[5] with an 0.99% higher Dice score than nnU-Net, while using 91.5% less parameters, 85.05% less FLOPs, and 31.94% less memory usage compared with nnU-Net. The proposed model is compared against nnU-Net,[5] SegResNet,[8] and HCMA-UNet.[9] All models use U-Net-inspired encoder–decoder architectures.[10,11] HCMA-Unet is part of a design of efficient models utilizing Mamba state space models (SSMs). They offer an efficient alternative to transformers that can capture long-range dependencies[12] but suffer from quadratic attention complexity.[12] Mamba SSMs are sequence models, known for their strong capability in handling long sequences.[13] U-Mamba achieves comparable performance to SOTA convolutional neural network (CNN)-based and transformer-based models while achieving linear computational complexity.[13] The HCMA-UNet study[9] claimed better performance on the MAMA-MIA dataset than U-Mamba and MambaClinix.[14] MambaClinix[14] extended the Mamba SSM design with a hierarchical gated convolutional network (HGCN) that is designed to mimic the attention mechanism of Transformers by a purely convolutional structure. The VMamba[15] model adds a visual SSM block (VSSB) with scan-based strategies (e.g., cross-scan) to convert spatial data into sequence-friendly formats that SSMs can operate on, effectively capturing global context in 2D while keeping linear cost. HCMA-UNet concatenates its axial self-attention (ASA) module to the VSSB, enabling the capturing of spatial dependencies in medical volumes while reducing the quadratic complexity to the axis.[9] SegMamba[16] completely removes the quadratic complexity by replacing the ASA with a tri-orientated spatial Mamba block, approximating the 3D attention in linear time.

In addition to segmentation, the last encoder block output of our proposed model is experimented with to predict the pathological complete response (pCR) to neoadjuvant chemotherapy (NAC). We applied an XGBoost model on the averaged pooled features of the last encoder block. In a study by Chen et al.,[17] an XGBoost model trained on antigen and other risk factors resulted in successful pCR prediction for locally advanced rectal cancer patients after NAC. Achieving a jump in ROC-AUC from 0.72 to 0.95 over their nomogram model. Unfortunately, our results were insufficient with a ROC-AUC of 0.542.

2 Methods

2.1 Proposed Model

Our proposed model is illustrated in Fig. 1. It follows the SegResNet[8] encoder–decoder structure and applies the same number of residual blocks as SegResNet.[8] An initial convolution increases the three-channel input to 32 channels while keeping the image dimensions identical. The encoder uses convolutions with a stride of 2 to downsample by a factor of 2 and simultaneously increases the number of channels by a factor of 2. Tumor segmentation requires global context to know which regions are plausible, and to compare the intensities within the MRI, an encoder is utilized. The end of the encoder has shape $256 \times 7 \times 24 \times 24$ for each input patch with shape $3 \times 56 \times 192 \times 192$. The number of patches k needed for inference is dependent on the input shape. Therefore, the extracted deep feature from our model used for pCR prediction is calculated by summing all k deep features together and dividing by k. The decoder applies pointwise convolutions for channel reduction, followed by interpolation for dimension upsampling. The decoder mostly helps in spatial precision and refined boundary segmentation. Before each ResBlock, the output of the encoder's ResBlock at the same spatial level is added to the decoder

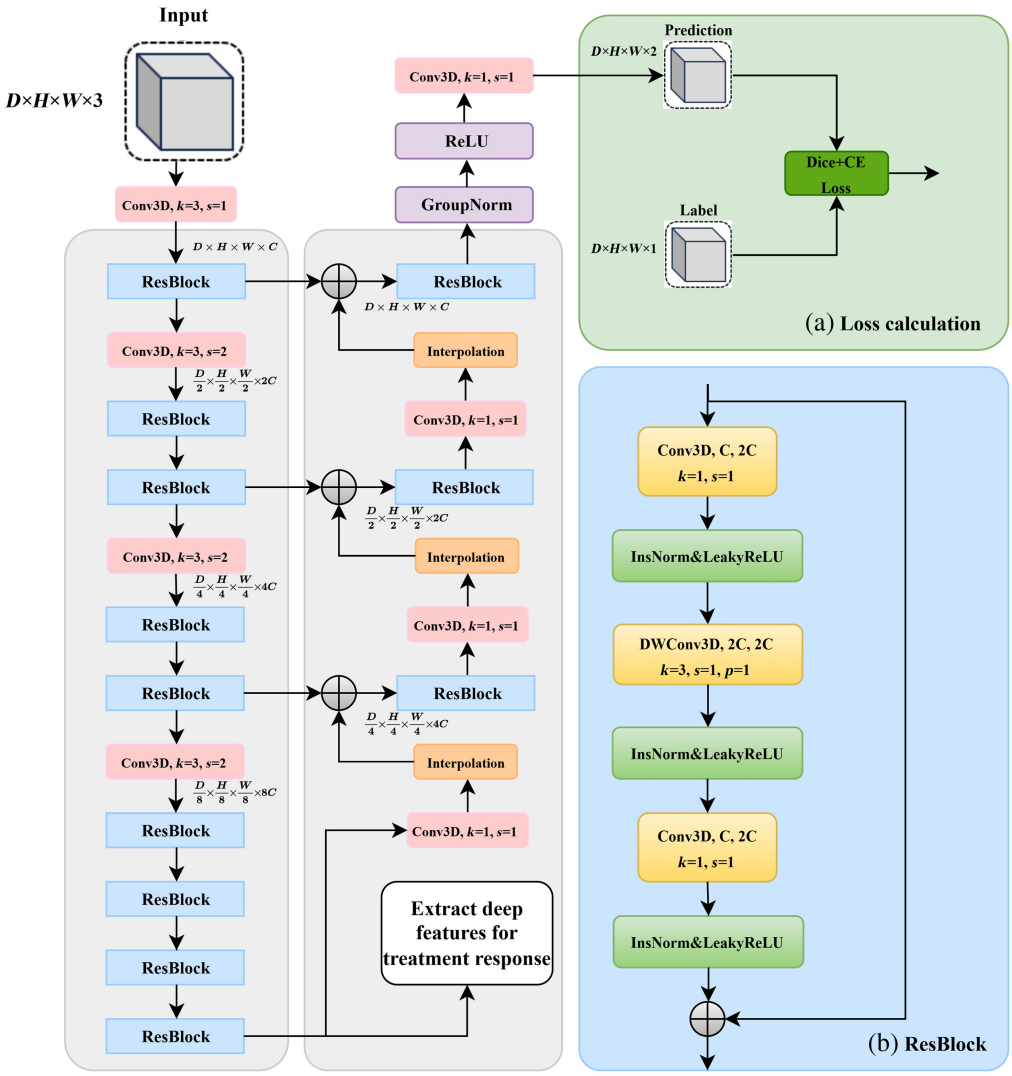


Fig. 1 Overview of the proposed model. (a) Loss computation module that combines the Dice[7] and cross-entropy[6] loss. (b) Our proposed ResBlock, inspired by HCMA-UNet,[9] the stride for all convolutions is 1 because the downsampling operations are performed in separate convolutions before each group of ResBlocks in the encoder.

features; this preserves encoder features that may have been lost deeper within the encoder. The output block applies GroupNorm over eight groups, followed by ReLU and a final convolution mapping the feature maps down from 32 to 2.

2.2 ResBlock

Our proposed lightweight residual block shown in Fig. 2 is inspired by HCMA-UNet's[9] ResBlock. The lightweight ResBlock reduces FLOPs by using small kernels ($1 \times 1 \times 1$) for regular convolutions and larger kernels ($3 \times 3 \times 3$) with depthwise convolutions. As our model performs convolutions with a stride of 2 before each group of ResBlocks in the encoder, all strides in

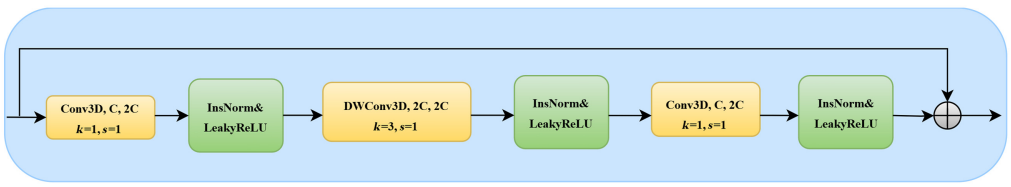


Fig. 2 Our proposed lightweight ResBlock, inspired by HCMA-UNet.[9]

the ResBlock itself are 1 as no further downsampling is required. The channels are expanded by a factor of 2 before entering the depthwise convolution. This type of convolution saves parameters compared with a traditional convolution while still utilizing a $3 \times 3 \times 3$ kernel to capture spatial context. Depthwise convolutions also do not apply all filters over all channels, but only one filter per channel. Therefore, there are fewer intermediate feature maps, and less memory is needed to store them. As demonstrated in Eqs. (1)–(4), our lightweight ResBlock scales its parameters by $C(4C + 54)$ with respect to the input channels. The original ResBlock in SegResNet[8] uses two $3 \times 3 \times 3$ convolutions, one such convolution scales by $27C^2$, shown in Eq. (5). Thus, the total amount of parameters in SegResNet's ResBlock is $54C^2$.

$$1 \times 1 \times 1 \text{ Conv3D Params} = C \cdot 2C \cdot 1 \cdot 1 \cdot 1 = 2C^2, \tag{1}$$

$$3 \times 3 \times 3 \text{ DWConv3D Params} = 2C \cdot 3 \cdot 3 \cdot 3 = 54C, \tag{2}$$

$$\text{Lightweight ResBlock Params} = 2 \cdot 2C^2 + 54C, \tag{3}$$

$$\text{Lightweight ResBlock Params} = C(4C + 54), \tag{4}$$

$$3 \times 3 \times 3 \text{ Conv3D Params} = C \cdot C \cdot 3 \cdot 3 \cdot 3 = 27C^2, \tag{5}$$

2.3 Training Loss

All models are trained with a combination of the Dice[7] and CE[6] losses. To compare our model to HCMA-UNet,[9] we had to remove the FRLoss component from that model. However, in Sec. 3.4, we performed an FRLoss ablation over all models, in which we include the loss alongside the Dice and CE losses. For this reason, we break down its three components—positive compactness loss, boundary-aware loss, and hard negative pixel mining loss.

- 1. Positive compactness loss: This loss promotes feature compactness by maximizing the similarity of individual lesion features with the mean lesion feature.[9] This loss can be helpful because lesion pixels can vary a lot internally, and this loss forces lesion features to be more similar and avoids scattered predictions. The mean lesion is the foreground center f_p, and $sim(\cdot)$ represents the cosine similarity,[9] N_{pos} denotes the total number of positive pixels, and $f_i \in R^C$ represents the feature vector of the i'th foreground pixel extracted from feature maps before the output block.[9]

$$f_p = \frac{1}{N_{\text{pos}}} \sum_{i=1}^{N_{\text{pos}}} f_i, \tag{6}$$

$$\mathcal{L}_{\text{pos}} = \frac{1}{N_{\text{pos}}} \sum_{i=1}^{N_{\text{pos}}} (1 - sim(f_i, f_p)), \tag{7}$$

- 2. Boundary-aware loss: This loss enhances boundary segmentation by calculating the similarity between the foreground center and the features in the boundary confusion region denoted as $\mathcal{M}$.[9] It discourages background features close to the border to be similar to the mean lesion feature, aiming to improve the detail in boundary classification. The boundary confusion region is formed by performing T_1 times dilation with a $3 \times 3 \times 3$ kernel on the positive pixel set (Ω_{pos}) and intersecting with the negative pixel set (Ω_{neg}).[9]

$$\mathcal{M} = \text{dilate}^{T_1}(\Omega_{\text{pos}}) \cup (\Omega_{\text{neg}}) \tag{8}$$

$$\mathcal{L}_{\text{boundary}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} ReLU(sim(f_i, f_p)) \tag{9}$$

- 3. Hard negative pixel mining loss: This loss penalizes negative regions with high similarity to the foreground center.[9] It searches for the top N negative pixels, which are most similar to the mean lesion feature. These pixels then resemble the hard negative pixels and are penalized more to help the model learn segmenting them as negative.

$$S = \{s_i = sim(f_i, f_p) | i \in \Omega_{\text{neg}}\} \tag{10}$$

$$\mathcal{N} = \text{dilate}^{T_2}(\text{top}_N(S)) \cap \Omega_{\text{neg}} \tag{11}$$

$$\mathcal{L}_{\text{neg}} = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} ReLU(sim(f_i, f_p)) \tag{12}$$

3 Dataset, Experiments, and Results

3.1 Dataset

The public MAMA-MIA dataset[1] consists of 1506 cases; the details of which are provided in Table 1. We divided the cases into 1204 train and 302 validation samples. This split is based on an even distribution of menopausal status, age, and breast density advised by the MAMA-MIA dataset and can be found on their synapse page. This split references the public train and validation split of the dataset, whereas the test set is kept private. Therefore, the results in this study correspond to the validation set, which is consistent with the methodology used in prior studies. Each case contains a pre-contrast phase and up to two to five post-contrast phases; this is illustrated in Fig. 3. As nnU-Net's preprocessor[5] requires a fixed number of input channels and some cases only have two post-contrast phases, we used the pre-contrast phase and the first two post-contrast phases per case to enforce a fixed three-channel input and to retain most of the sequences.

For the treatment response experiment, we used the same split. However, some cases do not contain a pCR label; these have been left out. The final split includes 1192 train and 299 test samples. In the train set, the negative:positive ratio is 3.32:1, for the test set, this ratio is 3.69:1. We also trained four different models, each trained on one subgroup in the MAMA-MIA dataset[1] (DUKE, ISPY1, ISPY2, NACT) and evaluated over each distinct group. The left table in Table 2 displays the number of pCR cases per subgroup in the MAMA-MIA dataset.[1] We used these data to train our models. A comparative study by F. Schwarzhans et al.[4] trained logistic regression models on the triple negative breast cancer (TNBC) cases, the number of pCR cases per subgroup used in their research is displayed in the second table in Table 2, and they did not use NACT due to the low amount of pCR cases.

Table 1 Properties of the dataset used in this work.[18]

		Total	1506
		Cases	%
Pathologic complete response (pCR)	Yes	440	29.2
	No	1051	69.8
	N/A	15	1.0
Age group	<40	317	21.0
	40 to 49	491	32.6
	50 to 59	465	30.9
	60 to 69	199	13.2
	>70	31	2.1
	N/A	3	0.2
Menopausal status	Pre	678	45.0
	Post	420	27.9
	N/A	408	27.1
Number of DCE-MRI phases	3	205	13.6
	4 to 5	316	21
	6	985	65.4

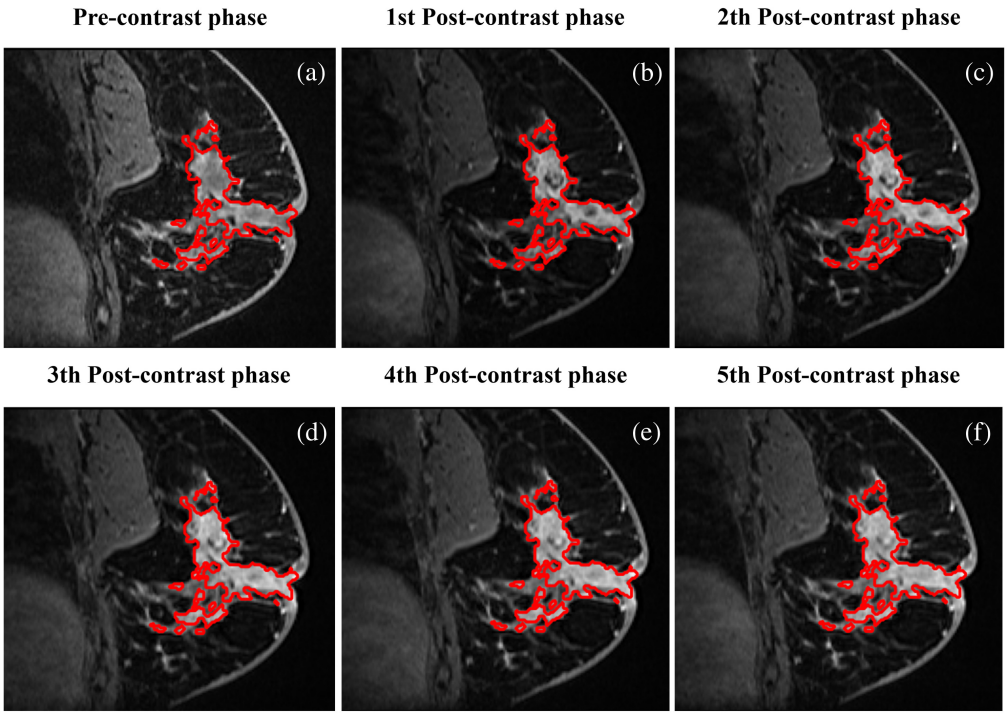


Fig. 3 Images of all DCE-MRI phases of sample ISPY2_415039. The expert tumor segmentation is indicated by the red line. (a) Pre-contrast phase, (b) first post-contrast phase, (c) second post-contrast phase, (d) third post-contrast phase, (e) fourth post-contrast phase, and (f) fifth post-contrast phase.

Table 2 Top: the number of cases used for training the XGBoost model on our proposed model's deep features. Bottom: the number of cases used in the comparative study.[4]

Subgroup	Total subjects	Non-pCR	pCR
DUKE	291	216	64
ISPY1	171	118	49
ISPY2	980	664	316
NACT	64	53	11

Subgroup	Total TNBC	Non-pCR	pCR
DUKE	82	59	23
ISPY1	43	28	15
ISPY2	358	217	141
NACT	11	8	3

3.2 Implementation Details

All experiments were conducted on an NVIDIA H100 GPU. Data preprocessing and augmentation followed the nnU-Net framework,[5] using its 2d_dummy augmentation. Models were trained with a patch size of $56 \times 192 \times 192$, anisotropic spacing of $2 \times 0.7 \times 0.7$ mm^3, batch size 2, for 1000 epochs. Training used a learning rate of 1e−4, poly-linear scheduler, Adam optimizer[19] with momentum 0.9, and weight decay of 1e−5; dropout was disabled. All models used cross-entropy, and Dice with equal weights of 1. The FRLoss ablation used a Dice, cross-entropy, and FRLoss combination with weights 1, 1, and 5, respectively. Post-processing

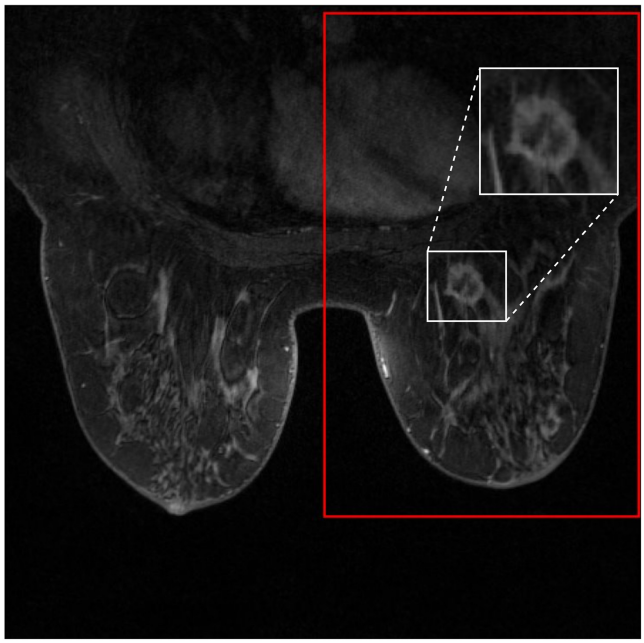


Fig. 4 Example of a cropping mask outlined in red. The tumor is specified in the zoomed-in region.

involved cropping predictions using MAMA-MIA masks[1] and filtering components smaller than 950 pixels. An example of the MAMA-MIA mask is presented in Fig. 4; these masks filter out the breast without the primary tumor. The segmentation performance is evaluated using Dice[7] and 95th percentile Hausdorff Distance (HD95).[20] The final segmentation score is the average over the Dice and (1 - NormHD95). NormHD95 is the HD95 normalized with a predetermined maximum of 150 mm. The pCR prediction performance is evaluated with the balanced accuracy, which is the average of the sensitivity and specificity. For empty predictions (~0.3% of cases), Dice is set to 0 and HD95 to 150 mm. Efficiency is reported in parameter count, giga floating point operations per second (GFLOPs), memory usage, and inference time. GFLOPs and memory are measured per patch using the fvcore library;[21] inference time is averaged per case over the validation set.

3.3 Result Comparison

As shown in Tables 3 and 4, our proposed model demonstrates significantly better Dice scores than nnU-Net ($p = 0.0374$) and HCMA-UNet ($p = 0.0000$), with a Dice score of 80.95%, with only 2.61 M parameters, 87 GFLOPs, and 1.658 GB memory usage. This is substantially lower than nnU-Net (30.70 M parameters, 555 GFLOPs, 2.436 GB memory) and SegResNet (18.79 M parameters, 576 GFLOPs, 1.905 GB memory). It uses less parameters and memory compared with HCMA-UNet (2.87 M parameters, 38 GFLOPs, 2.053 GB memory). Our model also outperforms nnU-Net in HD95 score ($p = 0.0373$). However, both nnU-Net and HCMA-UNet

Table 3 Comparison of different architectures for 3D breast tumor segmentation evaluated on the public MAMA-MIA dataset.[1]

Models	Segmentation score	Dice	HD95	MParams	GFLOPs	GB/ memory	Inference time (s)
nnU-Net	82.80	79.96	21.54	30.70	555	2.436	**2.58**
SegResNet	83.01	80.89	22.30	18.79	576	1.905	12.52
HCMA-UNet	81.74	79.04	23.34	2.87	**38**	2.053	5.96
Ours	**83.61**	**80.95**	**20.61**	**2.61**	87	**1.658**	10.83

Note: bold entries represent the best values in the column.

Table 4 Wilcoxon-signed-rank test[22] *p*-values for pairwise comparisons between models based on Dice and HD95 scores. Bold values indicate statistical significance at $\alpha = 0.05$.

Dice *p*-values	SegResNet	nnU-Net	HCMA-UNet
Ours	0.1355	**0.0374**	**0.0000**
SegResNet	–	0.8780	**0.0009**
nnU-Net		–	**0.0188**
HCMA-UNet			–
HD95 *p*-values	SegResNet	nnU-Net	HCMA-UNet
Ours	0.5131	**0.0373**	0.3704
SegResNet	–	0.3550	0.1692
nnU-Net		–	0.8085
HCMA-UNet			–

show faster inference speed, and HCMA-UNet operates on less GFLOPs. SegResNet achieves comparable performance. These results create a trade-off. For superior performance, our model and SegResNet[8] are the strongest models, with our model being the more parameter-efficient option. Even so, both still have the slowest inference speeds over all four models. To obtain the fastest inference speed, nnU-Net offers the best option. Last, HCMA-UNet can be used with low-parameter constraints and inference speed being a priority, if the user can afford a small drop in performance.

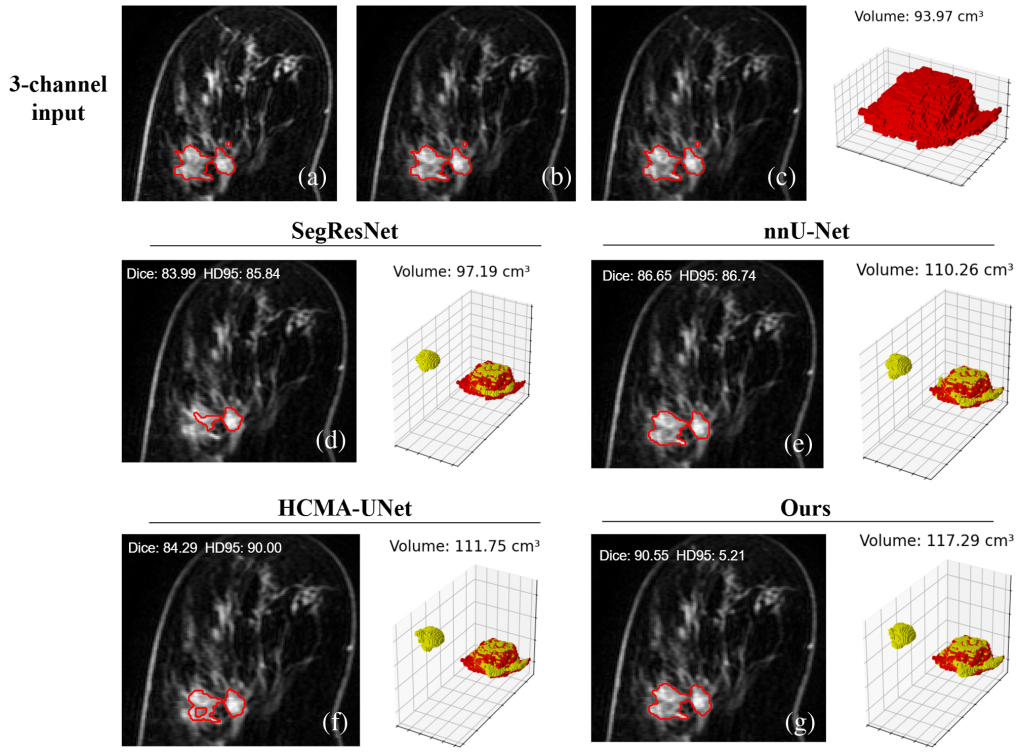


Fig. 5 Visual comparison for sample ISPY2_510543 shows ground truth and segmentations from nnU-Net (e),[5] SegResNet (d),[8] HCMA-UNet (f),[9] and our proposed model (g). Panels (a)–(c) display the three-channel input with red lines indicating ground truth. The ground truth 3D segmentation and volume are shown alongside. Each model's 2D and 3D segmentation views are also presented in yellow, with the ground truth in red.

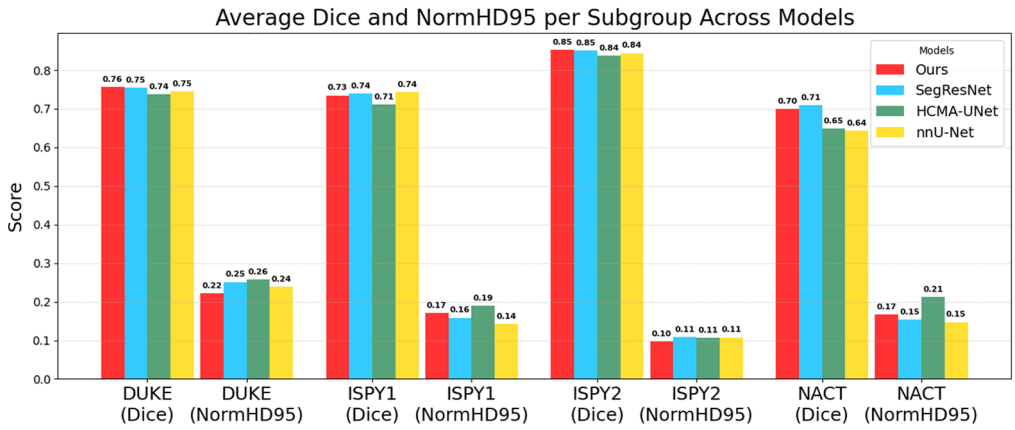


Fig. 6 Dice and NormHD95 scores per subgroup. The NormHD95 is the HD95 normalized by dividing by the predetermined maximum of 150 mm.

In Figs. 5(a)–5(c), an example of the three-channel input is given. Our model demonstrates a higher level of overlap and boundary segmentation accuracy. The volumes indicate a clear deviation from the target volume, this is caused by all models predicting a false positive mass in the same spot. The red target volume overlay in Figs. 5(d)–5(g) show that especially SegResNet is under-segmenting the tumor, which is clearly observable due to its lower volume compared with the other models. nnU-Net and HCMA-UNet also under-segmented this case, this led to the larger HD95 and lower Dice scores.

In Fig. 6, the Dice and normalized HD95 scores per dataset subgroup are illustrated. Our model achieves the highest Dice scores and lowest HD95 scores for the DUKE and ISPY2 subgroups.

3.4 FRLoss Ablation

In the HCMA-UNet study,[9] models were trained on the first post-contrast phase, and their proposed FRLoss improved segmentation. Similarly, we observed improved results using FRLoss with SegResNet[8] on a single-phase input (Table 5). In this ablation, we train all four models with the combination of the Dice, CE, and FRLoss losses, using the three-phase input. Using the results in Table 5, we can support that the addition of the FRLoss does improve performance

Table 5 Ablation over the addition of FRLoss to the Dice and cross-entropy losses for multiple numbers of input channels.

Models	Input channels	Segmentation score	Dice	HD95
SegResNet	1	80.89	78.71	25.41
SegResNet w/ FRLoss	1	82.46	79.14	21.34
SegResNet	3	83.01	80.89	22.30
SegResNet w/ FRLoss	3	**83.73**	80.48	**19.52**
nnU-Net	3	82.80	79.96	21.54
nnU-Net w/ FRLoss	3	82.99	79.98	21
HCMA-UNet	3	81.74	79.04	23.34
HCMA-UNet w/ FRLoss	3	82.64	79.47	21.30
Ours	3	83.61	**80.95**	20.61
Ours w/ FRLoss	3	83.26	80.37	20.79

Note: bold entries represent the best values in the column.

Table 6 Wilcoxon-signed-rank test[22] p-values for pairwise comparisons between models based on Dice and HD95 scores, with FRLoss. Bold values indicate statistical significance at $\alpha = 0.05$.

Dice p-values	Ours w/ FR	SegResNet w/ FR	nnU-Net w/ FR	HCMA-UNet w/ FR
Ours w/o FR	0.5208	0.6458	0.1678	**0.0001**
SegResNet w/o FR	0.2851	0.8004	0.8570	**0.0102**
nnU-Net w/o FR	0.3453	0.5725	0.2113	**0.0793**
HCMA-UNet w/o FR	**0.0000**	**0.0022**	**0.0077**	0.9421
Ours w/ FR	—	0.6096	0.5500	**0.0003**
SegResNet w/ FR	—	—	0.4532	**0.0025**
nnU-Net w/ FR	—		—	**0.0278**
HCMA-UNet w/ FR	—			—
HD95 p-values	Ours w/ FR	SegResNet w/ FR	nnU-Net w/ FR	HCMA-UNet w/ FR
Ours w/o FR	0.1545	0.2361	0.3996	0.9785
SegResNet w/o FR	0.1667	0.1459	0.6881	0.8871
nnU-Net w/o FR	**0.0196**	**0.0041**	0.4740	0.1813
HCMA-UNet w/o FR	**0.0134**	**0.0477**	0.3416	0.4931
Ours w/ FR	—	0.8901	0.2000	0.2844
SegResNet w/ FR	—	—	0.2788	0.1152
nnU-Net w/ FR	—		—	0.9910
HCMA-UNet w/ FR	—			—

for HCMA-UNet, as well as for nnU-Net and SegResNet, but not for our model. SegResNet's jump in segmentation performance surpasses the segmentation score of our proposed model without FRLoss by 0.12 ($p = 0.2361$ in HD95). This is a surprising result; it seems that larger models and mamba-based models may benefit from the addition of the extra loss, although not with a significant difference, as demonstrated in Table 6. These models may have vaster representational capacities that can utilize an extra loss signal. By contrast, our smaller model may not have this ability and may already have some kind of inherent regularization due to the smaller size. Our model achieves the highest Dice score over all configurations, and SegResNet achieves the lowest HD95 score. It is important to note that our model no longer significantly outperforms nnU-Net. Therefore, we claim comparable performance to nnU-Net and SegResNet and significantly better performance to HCMA-UNet.

3.5 Ablation Experiment with Our Model over Input Phases

As can be seen in Fig. 3, the tumor area sometimes only enhances after multiple phases. This indicates the presence of early- and late-enhancing tumors. Therefore, we tested multiple training configurations that experimented with a number of different input options; these results are shown in Table 7. This ablation made use of the extra FRLoss. Our final approach uses the three-channel input made up of the pre-contrast phase and the first and second post-contrast phases.

3.6 Network Depth Ablation

Our model uses a 1-2-2-4 encoder and 1-1-1 decoder with three down/upsampling steps. We compared this to a deeper 2-2-2-2-2-2 encoder and 2-2-2-2-2 decoder, both trained with FRLoss[9] on the first post-contrast phase. As shown in Table 8, the deeper model achieved a significantly higher Dice score ($p = 0.0175$), but the smaller model had a better HD95 ($p = 0.0208$). Due to the

Table 7 Ablation study over the configuration of DCE-MRI phases used during training and inference of our model.

Pre	1	2	Performance score	Dice	HD95	GFLOPS	GB/ memory	Inference time (s)
—	✓	—	81.96	78.82	22.35	**83**	**1.643**	**10.13**
—	—	✓	80.70	77.65	24.38	**83**	**1.643**	10.62
✓	✓	✓	**83.26**	**80.37**	**20.79**	87	1.658	10.83

Note: bold entries represent the best values in the column.

Table 8 Comparison of different residual block configurations in the encoder and decoder of our proposed model.

Models	Encoder–decoder configuration	Dice	HD95	MParams	GFLOPs	GB/ memory	Inference time (s)
Ours	[1,2,2,4]-[1,1,1]	78.82	**22.35**	2.61	87	1.658	10.83
Ours	[2,2,2,2,2,2]-[2,2,2,2,2]	**80.04**	22.57	33.89	124	2.376	17.09

Note: bold entries represent the best values in the column.

higher computational cost, the deeper model was not pursued further, though its Dice score (80.04) exceeded SegResNet's (79.14) in Table 5, suggesting future potential.

3.7 Treatment Response Prediction

The extracted deep features ($256 \times 7 \times 24 \times 24$) are average-pooled to a 256-dimensional vector. We trained XGBoost[23] on these vectors (1192 train, 299 validation samples), achieving 57.2% balanced accuracy and 0.542 receiver operating characteristic–area under curve (ROC AUC). The model used two estimators (max depth 5, learning rate 1). We also evaluated subgroup-specific models, comparing results to F. Schwarzhans et al.,[4] which used 107 radiomic features with logistic regression. In Fig. 7, it can be seen that our model shows better performance over ISPY2 when trained on ISPY1 and better performance over DUKE when trained on ISPY2. However, over all four models, it does not achieve the highest ROC AUC score for any subgroup, suggesting that the deep features may lack sufficient discriminative power likely due to information loss in the encoder or the restriction of using only the first three phases of the DCE-MRI. This restriction was applied due to the fixed input size requirement of the nnU-Net[5] framework. However, the majority of cases consisted of 4, 5, or 6 phases.

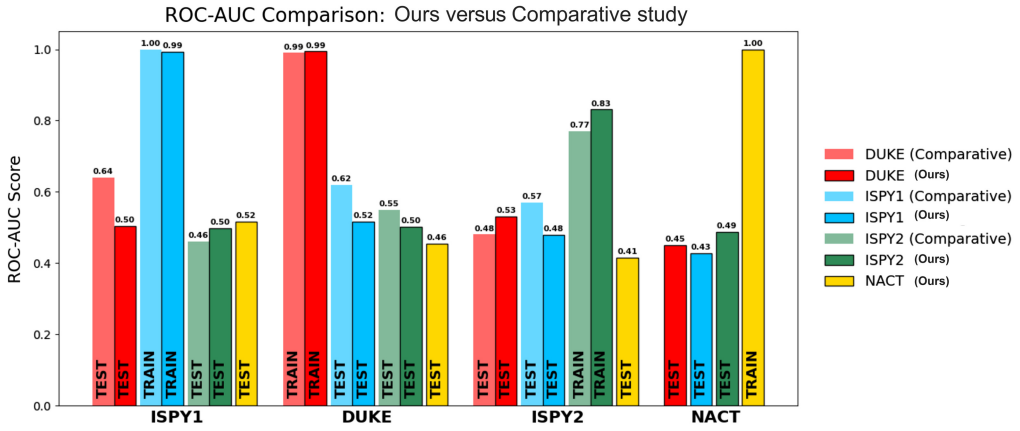


Fig. 7 x labels represent the subgroup used for training the model. The research of Schwarzhans et al.[4] represents the comparative study.

Table 9 pCR models evaluated over the public MAMA-MIA validation set.[1]

pCR models	ROC (%)	BAcc (%)	Sensitivity	Specifity	Precision (%)	NPV (%)	TP	TN	FP	FN
Majority class (=Negative)	50	50	0.0	1.0	0	72.9	0	218	0	81
Minority class (=Positive)	50	50	1.0	0.0	27.1	0	81	0	218	0
XGboost	54.2	57.2	0.31	0.83	30.9	76.5	25	182	36	56

Note: bold entries represent the best values in the column.

Later phases in DCE-MRIs may contain essential information due to the washout of the contrastive agent.[24] Ambramson et al.[24] showed that the percentage of voxels demonstrating washout kinetics posed as a discriminating factor between pCR and non-pCR patients, resulting in an ROC-AUC of 0.77 with a sample size of 30.

In Table 9, the positive predicted value (precision) and the negative predicted value (NPV) are given. For the naive minority classifier, the precision is 27.1%, which means that for the positive predicted group, 27.1% actually has a positive response to the chemotherapy. Our model achieves a precision of 30.9%, and this is a small improvement. Our model achieved an NPV of 76.5%; this is also a small improvement compared with the majority classifier with 72.9%. However, the model is not strong enough to be used for decision making.

4 Discussion

The ablation experiment over different input configurations results in big differences in performance. Thus, future research could focus on finding the best input sequences. In particular, research could focus on utilizing the entire DCE-MRI sequence instead of restricting the input channels to be of a fixed size. This approach could make the model agnostic to variable-length DCE-MRI sequences, most likely improving segmentation and pCR prediction performances due to the washout effect occurring in DCE-MRIs.

Although our model achieves the highest Dice score, SegResNet achieves a lower HD95 score with the addition of the FRLoss. Further research into accurate boundary segmentation through model architecture can potentially lead to boundary refinements. Although nnU-Net is the least efficient model in terms of parameters, GFLOPs, and memory usage, its inference speed is the fastest. This gap potentially decreases with smaller hardware. However, our proposed model could benefit from architectural designs or quantization to reduce inference time.

The FRLoss ablation experiment demonstrated that this loss may primarily benefit larger and mamba-based models. Meanwhile, our smaller architecture seems to benefit from its indirect regularization through its smaller size and does not need an extra loss. The network depth ablation suggests that the Dice score can be further optimized using deeper networks. However, these networks have a large number of parameters, memory usage, and inference time.

5 Conclusion

In this work, we propose a parameter-efficient CNN model for breast tumor segmentation in DCE-MRI, which integrates lightweight residual blocks into the SegResNet backbone. Our proposed model achieves state-of-the-art Dice performance and volume estimation while greatly reducing parameter count and peak memory usage. Experiments also demonstrate its strong capabilities in boundary segmentation. The model does not achieve competitive performance in pCR prediction likely due to the restriction of using only the first 3 DCE-MRI phases.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

Code and Data Availability

The code for this work can be found in the GitHub repository. GitHub: https://github.com/slndaniel/MamaMiaSubmission.

Disclaimer

The validation data used in this paper do not correspond to the private test set hosted by the MAMA-MIA challenge. Therefore, the results in this paper cannot be compared with those of the MAMA-MIA challenge leaderboard.

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